Modelling The Dependence Between Real Interest Rates And the Domestic Company Index In Botswana: A Copula Based Approach

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Declaration

I, Babusi Brian Rasenyai of student identity number: 202100936 do hereby declare that this mini dissertation is the result of my own investigation except where otherwise stated. I also declare that it has never been submitted previously as a whole or in part for any other degree at University of Botswana or any other institution.

Signature **Date**

This Mini dissertation has been submitted with my approval as a university supervisor and would certify that the requirements for the applicable Bachelor of Science rules and regulations have been fulfilled.

Signature **Date**

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List of Abbreviations

AIC: Akaike Information Criterion

BIC: Bayesian Information Criterion

BSE: Botswana Stock Exchange

CDF: Cumulative Distribution Function

CvM: Cramér-von Mises

DCC: Dynamic Conditional Correlation

DCI: Domestic Company Index

EDF: Empirical Distribution Function

GFC: Global Financial Crisis

IPO: Initial Public Offering

JB: Jarque-Bera

MLE: Maximum Likelihood Estimation

MPL: Maximum Pseudo-Likelihood

IFM: Inference Functions for Margins

PML: Pseudo Maximum Likelihood

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ABSTRACT

The purpose of this study is to model the relationship between real interest rates and stock market performance, specifically the Domestic Company Index (DCI), using a bivariate copula model. The study applies various bivariate copula models to assess the dependence structure between these two variables, determine their relationship, and identify the best model for representing the data. The analysis utilizes DCI data and interest rates data from January 2013 to December 2024. The Frank copula model revealed a weak negative dependence between interest rates and the DCI, indicating a subtle inverse relationship. Given the limited literature on this topic in Botswana, this study offers a valuable contribution by providing insights into the dynamic relationship between these variables in a local context. Copula models outperform traditional linear approaches in capturing nonlinear relationships and tail dependencies. Future research could enhance model robustness by incorporating additional economic variables and employing goodness-of-fit tests, such as the Cramér-von Mises test, to further validate the copula models.

Keywords: Bivariate Copula, Interest Rates, Domestic Company Index, Frank Copula, Dependence Modeling.

CHAPTER 1

INTRODUCTION

1.1 Background Of The Study

The economy of Botswana has experienced a multitude of blows as of late. There has been a dramatic increase in commodity prices, which has alerted the central bank. The rising inflation causes the central bank to tweak interest rates to mitigate the impacts of inflation. However, this prompts the question of how this affects the stock market and its performance. Although the stock market is affected by a lot of different factors it is worth looking into interest rates. It is important to explore the recent relationship between the two particularly in Botswana, by extracting the dependence structure.

We may define the rate of interest as the per cent of premium paid on money at one date in terms of money to be in hand one year later (Fisher 1930). Essentially, the interest rate is the cost of borrowing money. According to the Monetary policy report by the Bank of Botswana (2023), global economic performance and sentiment were adversely affected by the ongoing Russia Ukraine war, global monetary policy tightening and lingering effects of the COVID-19 pandemic, particularly the zero-COVID policies and lockdowns in China. In addition to the adverse impact on global economic activity, the war resulted in significantly elevated commodity prices and, therefore, inflationary pressures. In response, there was a generalized increase in policy interest rates and, hence, tightening of financial conditions. This shows there has been some increase in real interest rates recently. Focus will be on real interest rates.

According to Basvi (2024) a corporate or mutual entity that was founded for the purpose of exchanging investment instruments such as company stocks, securities, and derivatives, is known as an organized stock exchange. Here, investment tools are listed and traded under the supervision of nationally enacted laws and regulations. The stock market is an important economic indicator, it serves as a playground for investors. Although there are other factors that may affect the performance of the stock market, interest rate volatility leads to the need for investigation. The area of the stock market this study is going to focus on is the domestic company index (DCI). This DCI provides an extensive idea on the domestic companies listed on the Botswana stock exchange. It is important to investigate how changing interest rates affect the DCI as it is focused on domestic

companies in the country, one may say it is the most appropriate measure of the country's economic health. A rise in the DCI may suggest that the country is doing well economically. It may be appropriate to assume that the opposite is true, that is if the DCI is declining it may mean that the economic health of the country is poor.

Generally, the spending power of consumers and businesses is affected greatly by real interest rates. Since real interest rates are the cost of borrowing, when they rise, we can assume that investors have reduced spending power and this can lead to lower investment which negatively impacts corporate earnings, and the converse may be true. Thus, it is important to understand the relationship between these two commodities. It is important to note that according to literature this general assumption is not always the case, this relationship is affected by a numerous number of factors, so it varies throughout different world economies. The results contrast with the findings found throughout academic literature and indicate that the broadly accepted notion of a negative correlation between interest rate changes and stock returns is not valid for all economic regions(Rohner et al. 2021). This study shows that the general assumption that interest rates and stock market performance share an inverse relationship is not always the truth.

This study will explore how these two commodities have been interacting from 2013-2024. In this time period major economic events have taken place which will present a clearer picture of the relationship between these two variables. As mentioned in the beginning of the study, the effects of the COVID-19 pandemic resulted in an explosion in inflation and as a result the central bank spiked bank rates (Bank of Botswana monetary policy 2023). In terms of the stock market the covid 19 pandemic had an impact on the stock returns in Botswana. According to Ofori-Boateng et al. (2022) their results suggest that COVID-19 outbreak has a negative and statistically significant effect on stock market performance. Makatjane (2024), explained that major financial crises, like the Global Financial crisis (GFC) OF 2007-2008, the European Sovereign Debt Crisis (ESDC) of 2010-2011, and the 2019-2021 COVID- 19, typically affect the entire economy, causing sever downturns and recessions. It is also important to consider the stock market performance during this period. According to the annual Botswana stock exchange report (2022) At global level, 2022 was undoubtedly a challenging period as economies were trying to recover from the pandemic amid bleak growth prospects. Global Initial Public Offerings (IPOs) fell by 45% following a record year in 2021, according to EY's Global IPO Trends 2022 Report. They

further explain that under the high-interest rate environment, investors reallocated their portfolios to less risky assets leading to depressed company valuations and stock market indices, and this affected many prospective IPO plans. In light of all these economic events it proves to be rather important to study these commodities from during this timeframe.

This study will explore the different copula functions to measure dependence. According to Basvi (2024), a copula is a multivariate cumulative distribution function that is uniform across all marginal distributions on an n-dimensional cube when defined on [0, 1]. It describes the relationship between multiple variables, making it a useful tool for modeling dependences among them, (Basvi, 2024). According to Dewick and Liu (2022), Copula modelling is a popular tool in analyzing the dependencies between variables. They further explain that copula modelling allows the investigation of tail dependencies, which is of particular interest in risk and survival applications. Researchers have found that modelling dependence using copula functions is a very viable method particularly in financial data which could be characterized by extreme values, heavy tails and asymmetry. Makatjane (2024), the simple linear correlation, despite being widely used is unable to capture crucial tail behavior of a joint probability distribution. They further explain that in a multivariate environment, especially in light of recent crises episodes, modelling the tail dependence and asymmetric dependence between pairs of assets has been becoming increasingly important. The copula is adapted to modelling such data.

In terms of the economy, the stock market is at the heart of economic performance. The economy of Botswana is in dire need of diversification, considering how Botswana's diamond has been performing as of late. Therefore, this paper may serve as a guide to investors as to how to maneuver the stock market and also advise policy makers to limit the increase of interest rates. As a result, there will be a healthy increase in the general performance of the stock market.

1.2 Problem Statement

Exploring the relationship between interest rates and stock market performance is of paramount importance. Economic challenges like plummeting diamond sales and post-covid inflation caused an increase in real interest rates which may have a negative impact on the stock market particularly the DCI. Understanding the dependence structure between these real interest rates and stock market performance (DCI) is crucial for investors and policy makers.

As mentioned above financial data in Botswana often exhibits fluctuations. These fluctuations could make the data non-linear, asymmetric and heavy tailed. This means that traditional methods of modelling dependence like simple linear correlation may fail and hence it calls for other methods. This study is going to use a copula-based approach which is tailored to modelling such data.

This study also seeks to fill a gap in existing literature. Despite the importance of this relationship, and the advantages of using copulas, very few studies like this exist. Based on my investigations I failed to find any studies that used copulas to model the relationship between real interest rates and stock market performance particularly the DCI. Hence highlighting the importance of this study.

1.2.1 Research Objectives

The main objective of this study is to model the dependence of real interest rates and stock market (DCI) performance applying a copula function approach.

Specific outcomes

-to investigate the relationship between real interest rate and domestic company index using the Kendall's tau statistic.

-Fit various copulas to the data.

- to use AIC and BIC model selection criteria to select the best copula.

1.3Significance Of The Study

This study will contribute to the limited literature on real interest rates and stock market performance (DCI). Upon investigation I found out there is very little existing literature on this particular topic. Therefore, studying this relationship will serve as a basis for other scholars who may want to delve into Botswana's stock market performance. This paper could also help in regulation of interest rates by the central bank understanding the adverse effects of the fluctuations of this economic entity on the general economy of Botswana. Furthermore, should there be a significant relationship between interest rates and stock performance this paper could advise potential investors on the right times to want to invest in stock market performance.

From a statistical point of view copula theory is an under-researched mathematical tool as most statisticians are quite stuck on traditional methods of data analysis like regression. Copula functions are a great modelling tool, particularly for financial data which could be non-normal and asymmetric. Modelling relationships using this tool could provide more precise results that could contribute to how people approach modelling financial data. It could lead to more accurate conclusions.

1.4 Organization Of the Study

This study contains five chapters which are presented as follows:

Chapter 1: explains the introduction and general motivation of the study, the background and significance of the study. The background of the general relationship between interest rates and stock market performance, the problem statement of the study, and finally the study objectives.

Chapter 2: critically analysis other studies that dealt with copulas. From the methodologies, to the results, explaining the importance and limitations of different methods that are used in copulas.

Chapter 3: consists of the methodologies that will be used in this study.

Chapter 4: entails the research data analysis and interpretation of research findings.

Chapter 5: this section has the conclusions and recommendations of the study.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This section of the report will critically analyze existing literature of copula financial modelling. Emphasis will be on exploring how different researchers and statisticians used this mathematical tool to model real interest rates and stock market performance and how their predictions panned out. Different conclusions, varying hypotheses and methodologies will be examined and as a result it will determine how this particular study will contribute to gaps existing in this field.

The relationship between interest rates and stock market performance remains an issue of concern throughout various world economies and hence should be investigated. A study by Rohner et al. (2021) claims that in most industrialized nations sustained demographic shifts, global excess savings and a slowdown in productivity development have contributed to a global decrease in real interest rates over the previous 20 years. Their study concluded that an increase in interest rates favors the development of stock returns in the long run. The relationship, however, also implies that if there was a decrease in interest rates, stock markets are on average losing value Rohner et al. (2021). It however shows that this relationship is presented differently in different world economies. It is worth noting that this study was analyzed using a simple regression model. Mbulawa (2015), exchange rates, treasury bill rates and deposit rates had a negative relationship while bank rate had a positive rate had a positive relationship with stock market performance during the period of low inflation. The afore study was also conducted using regression analysis. With copula modelling, capturing tail dependence is an advantage over regression models therefore my study will seek to contribute to literature in such ways, also can capture dependence of non-normal data.

Authors frequently apply the gaussian copula to modelling dependence. According Dewick and Liu (2022), blame is squarely put in using the Gaussian copula, by financial traders in selling vast quantities of new securities and expanding financial markets to unimaginable levels. The author highlights that the Gaussian copula's inability to account for tail dependence limits its use in estimating relationships in housing price movements that can lead to model miss-specification, which is what happened in the financial crash of 2007 and 2008. A properly specified model may have resulted in another copula shape apart from the Gaussian copula. The article highlights just

how dangerous it is to fail to fit the right copula model to different sets of data. It explores how analysts may have contributed to the financial crash of 2008 and 2007. Furman (2017) however choosing the right copula is not an easy task, and the temptation to prefer a tractable rather than a meaningful candidate from the encompassing copula toolbox is strong. In that study the author is sharing the same sentiments as myself and other authors, the main drawback of copula studies is the need to run to the gaussian copula which might not fit in some cases particularly in data that is non-normal. This study will avoid this problem by applying different model selection methods to ensure that the best model is selected. It will also take into account for normality of the variables or lack thereof. Some copulas like the gaussian are normal in nature and hence will fail to fit non normal data. By exploring normality, employing different copulas and using different selection criteria the problem of model misspecification will be avoided.

Copula depend on the marginals and parameter estimation methods. It is worth noting that in most cases researchers assume a specific distributional form for each marginal, then apply methods like the MLE to extract the parameters of the copula. Afzal et al (2023), this study used the pseudo observations of the standardized residuals from the fitted DCC model to estimate the marginals and then estimate the copula parameters using MLE. Their study was also modelling stock performance. Kamisan et.al (2024), explained that MLE is a parametric technique that focuses on determining the parameter values that maximize the likelihood function, based on the assumption that the data samples comes from a population that perfectly matches the model parameters. Kamisan et.al (2024) further explain that, this method is dependent on the analyst's accuracy in assuming the distributions for the marginals. It is rather risky particularly in financial analysis, where wrong results could lead to losses and tanks in the stock market. Ali et.al (2024), We found that under correct specification of copula and marginals, the ML method performs better than IFM and PML with respect to estimated variance, mean-squared error, and relative efficiency. However, in case of misspecified scenario for either copulas or marginals, PML performs better than ML and IFM, Ali et.al (2024). This study will explore an PML method to extract the copula parameters and will use the empirical distribution functions to fit the marginals. Doing this, avoids issues of wrong assumptions of the distributions which could lead to model misspecification and wrong results overall. Moreover, it isn't particularly popular so my study will explore this method to fill that gap in literature.

For model selection various copula studies use different methods to select the best fitting model. Many authors employ goodness of fit tests like the Cramer von mises to select the best fitting copula, which is an important exercise as stated above. Hassane et al. (2021), in each case, we will base ourselves on the Cramer- von Mises statistic for the choice of the best copula. According to Toprak et.al (2024), the CvM test is a method that measures the fit between two distributions by evaluating the sum of squares of the differences between the observed and theoretical distribution functions. Toprak et.al (2024) further mention that this test measures the fit of the model to the data by analyzing the differences between the observed and expected distributions from the model. If the probability distribution of the population is unknown, it may be difficult to accurately represent the population with a traditional probability distribution. However, the basis of this study is that distribution assumption is a taxing and risky process. We can assume that the data may be non-normal, therefore fitting different distributions to the data and testing for the best fitting distribution can be cumbersome. This study will apply model selection criteria like the AIC AND BIC to curb this problem. Should we apply the CvM test, the accuracy of the test is fully dependent on the accuracy of the assumption of the marginal distributions. Wrong assumptions could lead to wrong conclusions invalidating the results of such tests.

2.2.1Concluding Remarks

It is worth mentioning that literature on this particular topic is quite under researched. Extracting dependence between real interest rates and stock market performance is often an overlooked relationship. Moreover, copula studies aren't that popular in literature in Botswana. Making this study rather important.

In conclusion this work will bridge the gap existing in our literature today. Not only will it provide further studies with a fundamental foundation. It will work as a guideline for other researchers to develop specific copula models that will seek to develop how analysis is performed in the country and continent. Furthermore, upon reaching meaningful conclusions potential investors may have a guideline on how to maneuver the stock market.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This methodology will have a few sections, namely, Data description and source, estimation of marginals of each of the variables (real interest rates and stock market performance (DCI)), Sklar's copula theory, general mathematical definition of copulas, copula families for modelling dependencies and relationships, the methods of parameter estimation, dependence measures and model selection criteria.

3.2 Data Description And Data Analysis Framework

This paper serves to investigate the relationship between real interest rates (PRIME) and stock market performance (domestic company index, log returns) in Botswana specifically. The study will start from the beginning of 2013 till the end of 2024. The reason being that the stock market takes a while to react to changes in interest rates, further interest rates response to inflationary pressures also takes some time, therefore it was important to follow the relationship from this time to reach the most accurate conclusion. The sample of the study is monthly domestic consumer indices and monthly real interest rate, a total of 282 observations were used in this study. It is important to investigate monthly real interest rates and monthly DCI returns. Although most studies that involve the stock market always model data based on daily market prices, it was considered that doing so will provide very unclear results due to market prices taking some time to react to changes in interest rates. The sources of data include Botswana stock exchange (for domestic company index data) and Bank of Botswana (for real interest rate data).

3.2.1 Basic definition

To analyze stock data it is important to compute the log returns. This is consistent with other copula studies that were investigating the same. Log returns stabilize variance, capture relative returns and they also show heavy tails which is particularly important for copula studies. Foresto (2024),

we transform each of the indices into continuously compounded returns (Y_t) ,taking the log-differences of their value(P_t) over the interval [t-1,t].

Below is the equation used to compute log returns

$$Y_t = \log\left(\frac{P_t}{P_{t-1}}\right) \qquad (3.1)$$

3.3Descriptive Statistics

Measures Of Skewness Kurtosis And Normality Tests

Although the marginals can be estimated non parametrically it is still important to understand if the data is normal and linear. Some copula depend on normality of the data, therefore understanding if the data is normally distributed can help to pick the best copula to model the dependence. To understand how the data is we compute the kurtosis and the skewness statistics as well as the Jarque-Bera tests for normality. Hatem (2022) the normality of the data can be checked also according to skewness and kurtosis.

3.3.1Kurtosis

Hatem (2022), explain that kurtosis compares the dataset distribution to the normal distribution. Data with highly or positive kurtosis are those who tend to have heavy tails or outliers verses low or negative kurtosis light tails with lack of outliers. Apart from normality this cements how important this statistic is for copula tests. Copulas are especially skilled with dealing with asymmetric data and heavy tailed data to extract the tail dependence. Other copula studies indicated that the use of these statistics can also help to pre-identify the copula model that can best fit the data. Becker et.al(2021) we therefore also assess different nonnormality tests and simple moment measures, like skewness and kurtosis, to identify situations which support the reliable usage of Gaussian copula approach. This is particularly important as we have established that

picking the right copula could be incredibly important especially if they are used to forecast and potentially mitigate financial risks. Below is the kurtosis formula as explained by Hatem (2022)

$$a_4 = \sum \frac{(X_i - \bar{X})^4}{ns^4} \qquad (3.2)$$

Where

- X_i represents a data point in the dataset.
- $\bar{X}$ represents the mean
- n is the sample size
- s is the standard deviation

- *Interpretation (standard kurtosis)*
- -if the kurtosis is close to 0, then a normal distribution is often assumed.
- -if the kurtosis is less than zero, then the distribution has light tails.
- -if the kurtosis is greater than zero, then the distribution has heavier tails.

3.3.2 Skewness

According to Hatem (2022), skewness assesses the extent to which a variables distribution is symmetrical. Symmetry is particularly important in this study primarily because financial data is most times asymmetrical which highlights the necessity of using copula models. According to Falkentrom (2023), specifically, we were interested in the degrees of skewness in X required for adequate performance of the copula method. Other forms of non-normality may also be used; we chose skewness as an example of a common form of non-normality. Hatem (2022) has illustrated the skewness formula and its interpretation.

$$a_3 = \sum \frac{(X_i - \bar{X})^3}{ns^3} \qquad (3.3)$$

Where

- X_i represents a data point in the dataset.

-$\bar{X}$ represents the mean

- n is the sample size

- s is the standard deviation

Interpretation

- If the skewness is between -0.5 and 0.5 the data are fairly symmetrical.
- If the skewness is between -1 and -0.5 or between 0.5 and 1 the data are moderately skewed.
- If the skewness is less than -1 or greater than 1 , the data are highly skewed.

3.3.3 Jarque-Bera

To confirm the normality of the datasets, we applied the JB test to truly find out if the data is normally distributed or not. Aslam et.al (2021) the test is based on the null hypotheses there is no difference between the data under study and the normal distribution versus the alternative hypothesis. The test statistic of the JB test is a function of skewness and kurtosis. Suppose S,K and n are the sample skewness, kurtosis and the sample size for a data set then the statistic used for JB test under classical statistics is defined as;

$$JB = \frac{n}{6}[s^2 + \frac{1}{4}(K-3)^2] \qquad (3.4)$$

Where

s = sample skewness

k= kurtosis

n= sample size

3.3.4 Scatter Diagrams

A simple yet effective way of visualizing data is by a scatter diagram. A diagram that will seek to show the general relationship between interest rates and domestic company index. In a copula study, scatter plots can be an important tool in deciding on which copulas to focus on and furthermore the general direction of the study. The scatter diagram will tell whether the data is linear or nonlinear, the type of dependence and the asymmetry of dependence. The scatter diagram will be on a simple x,y plane.

3.3.5Quantile-Quantile Plots

Yang et.al (2021),a Q-Q plot is a scatterplot created by plotting two sets of quantiles- one is the data quantile and the other is the theoretical distribution quantile-against one another. The Q-Q plot is particularly important as is investigates the normality of the variables under study, real interest rates and the log returns of the DCI. They also serve to investigate if any of the variables are tail heavy which is of importance in this study. Dewick and Liu (2022), a graphical method to determine if any tail dependence is present can be represented in a Q-Q plot of the two standardized residuals for residuals for crude oil and natural gas.

3.4Copulas

3.4.1Sklars Theorem

To understand copula functions there is an important mathematical theory that needs to be considered. According to Nelson (1999), Sklar's theorem elucidates the role that copulas play in the relationship between multivariate distribution functions and their univariate margins. Therefore, it is relevant to describe it, as it is the foundation to the modelling that will be used in this research.

Theorem:**Sklar's Theorem**.

Let H be a joint distribution function with margins F and G. Then there exists a copula C such that for all x,y in $\bar{R}$,

H(x,y) = C(F(x),G(y))

If F and G are continuous, then C is unique; otherwise, C is uniquely determined on RanF x RanG. Conversely, if C is a copula and F and G are distribution junctions, then the junction H defined by (2.3.1) is a joint distribution function with margins F and G.

Mathematical Definition of Copulas

According to Israel (2018)

Given a random vector $X_1, X_{2,\ldots,} X_p$ its marginal cumulative distribution functions are $c(u_1, u_{2,\ldots,} u_p)$= $P[U_1 \leq u_{1,} U_2 \leq u_{2,\ldots\ldots\ldots,} U_p \leq u_{p,}]$

$$F_i(x) = P[X_i \leq x] \quad (3.5)$$

By applying the probability integral transformation to each component, the marginal distributions of $(U_1, U_{2,\ldots\ldots,} U_p)$=$(F_1(X_1), F_2(X_2), \ldots\ldots, F_P(X_p)$ are uniform. He proceeds to explain that the copula of $X_1, X_{2,\ldots\ldots,} X_p$ is defined as the joint cumulative distribution function of $U_1, U_{2,\ldots\ldots,} U_p$ for which the marginal distribution of each variable is uniform as U(0,1).

<u>Variables Used In This Study</u>

For the purposes of this study let ;

X=REAL INTEREST RATES (Prime %)

Y=DCI (Log returns)

Sample Size:

N=282 data points

3.4.2 Family Of Copulas

There are a variety of copulas that exist, each capable of modelling different datasets based on the characteristics of the data. This section will explore a few of these copulas and seek to show how each of them work and what they are best suited for. There are elliptical copulas like the gaussian and student's t which are designed to model linear dependencies. According to Liang (2023), these copulas induce a dependence structure of elliptical distributions, the same way Gaussian Copulas describe the dependence in multivariate Gaussian distributions. There are also Archimedean copulas like the frank and the clayton which are useful for modelling both positive and negative dependence. Hasan et al. (2021), explains that an important class of copulas are the Archimedean copulas, due to their ease of construction and ability to represent different tail dependencies. This makes them particularly important with modelling non linear and nonnormal data. Hence this study will use the Gaussian, Student's t, Frank and the Plackett copulas.

Gaussian

This is one of the copulas that have been used for analysis and dependence modelling. According to Joe (2015), this is the copula used in classical multivariate statistics after individual variables have been transformed to Gaussian.

Bivariate Gaussian Copula

When d = 2, the argument of the bivariate cdf is replaced by ρ ∈ [−1, 1], leading to:

$$c(u,v;\rho) = \Phi(\,\Phi^{-1}(u), \Phi^{-1}(V);\rho) \tag{3.6}$$

, 0<u,v<1

The conditional distribution is

$$c_{2|1}(u|v;\rho) = \Phi\frac{\left(\Phi^{-1}(V)-\rho\Phi^{-1}(u)\right)}{\sqrt{1-\rho^2}} \tag{3.6.1}$$

The copula density for -1< $\rho < 1$

$$c(u,v;\rho) = \frac{\frac{\phi_2\left(\Phi^{-1}(u),\Phi^{-1}(V);\rho\right)}{\phi(\Phi^{-1}(u))\phi(\Phi^{-1}(V))}=(1-\rho^2)^{-\frac{1}{2}}\exp\left\{-\frac{1}{2}(x^2+y^2-2\rho xy\right))}{(1-\rho^2)\}\exp\{\frac{1}{2}(x^2+y^2)\}} \tag{3.6.2}$$

With $x = \Phi^{-1}(u), y = \Phi^{-1}(v) and\ 0 < u, v < 1$

Where:

- x and y represent the variables of interest being the domestic company index and interest rates.

- u and v represent the marginals of the variables of interest.

- ρ this is the correlation coefficient, which lies in the range [-1,1]. It measures the linear dependence between the two random variables.

- Φ this denotes the CDF of the standard normal distribution. It is used to transform the uniform marginals (u,v) into standard normal variates (x,y).

-Φ^{-1} this denotes the inverse of the CDF of the standard normal distribution. It transforms the standard normal variates (x,y) back into the uniform marginals.

Plackett

The next section is the placket copula which could also be applicable to bivariate cases. According to Joe (2015) the idea of Plackett (1965) is a continuous bivariate distribution. He further explains that this copula is convenient for discretizing two continuous variables to binary so that the measure of dependence is constant.

$$For\ 0 \leq \delta < \infty\ the\ copula\ cdf\ is:$$

$$c(u,v;\ \delta) = \frac{1}{2}\ \eta^{-1}\{1 +\ \eta(\mathrm{u}+\mathrm{v}) - [(\mathrm{u}+\mathrm{u}))^2 - 4\ \delta\ \eta uv]^{\frac{1}{2}}\}, 0 \leq u, v \leq 1 \tag{3.7}$$

Where $\eta = \delta - 1$ the conditional distribution is

$$c(u,v;\delta) = \frac{1}{2} - \frac{1}{2}\ \frac{[\eta u + 1 - (\eta+2)v]}{[1+\ \eta(\mathrm{u}+\mathrm{v}))^2 - 4\delta\eta uv]^{\frac{1}{2}}} \tag{3.7.1}$$

The copula density is

$$c(u,v;\delta) = \frac{\delta[1+\eta(u+v-2uv)]}{[1+\ \eta(\mathrm{u}+\mathrm{v}))^2 - 4\delta\eta uv]^{\frac{3}{2}}},\ 0 < u, v < 1. \tag{3.7.2}$$

Where;

- δ is the association parameter, which determines the strength and the type of dependence between two random variables.
- $\eta(eta)$ the transformation of the association parameter, used to simplify the expressions in the copula function.

Frank

According toe Joe (2015), Frank (1979) obtained the bivariate copula function, but it was further studied by Nelson(1986) and Genest(1987). Joe (2015), explains that the bivariate frank is useful as a copula family with closed form CDF, and having both negative and positive dependence from countermonotonic to comonotonic. In conclusion he states that because of tail quadrant

independence, the bivariate Frank copula would seldom be a good choice, especially when tail dependence of variables is desired.

For $-\infty < \delta < \infty, the\ copula\ cdf\ is:$

$$c(u,v;\delta) = -\delta^{-1}\log\left(\frac{1-e^{-\delta}-(1-e^{-\delta u})(1-e^{-\delta v})}{1-e^{-\delta}}\right),\ 0 \le u,v \le 1 \tag{3.8}$$

The density is

$$c(u,v;\delta) = \frac{\delta(1-e^{-\delta})e^{-\delta(u+v)}}{[1-e^{-\delta}-(1-e^{-\delta u})(1-e^{-\delta v})]^2} -, 0 \le u,v \le 1 \tag{3.8.1}$$

The conditional cdf and its inverse are:

$$c_{2|1}(v|u;\delta) = e^{-\delta u}\left[\left(1-e^{-\delta}\right)\left(1-e^{-\delta v}\right)^{-1} - \left(1-e^{-\delta u}\right)\right]^{-1} \tag{3.8.2}$$

$$c_{2|1}{}^{-1}(p|u;\ \delta) = -\delta^{-1}\log\{\ 1-(1-e^{-\delta})/(p^{-1}-1)e^{-\delta u}+1]\} \tag{3.8.3}$$

Where:

- δ is the association parameter which determines the strength and type of dependence of the variables.
- (u,v) the marginals of the variables.
- p is the probability of the variable being a certain value.

Student's T

This copula is an elliptical one called the t-copula.

The form of t-copula is obtained by,

$$C^t_{v,d}(\widehat{U}_1,\ldots.,\widehat{U}_d) \coloneqq t_d\left(t_d{}^{-1}\ \ (u_1),\ldots.,t_d{}^{-d}\ \ (u_d)\right) \tag{3.7}$$

Where t_d is the joint distribution and thet_v is the univariate standard t-distribution with degree of freedom v.

3.5.1Estimating The Marginals

To model data using a copula function it is imperative to first estimate the marginals, those marginals will be transformed to a uniform joint CDF. The method that will be employed to estimate the marginals will be a non-parametric empirical distribution function. According to Ayari and Boutahar(2021), nonparametric approach makes few or no assumptions about the data. Using the empirical distribution function represents an alternative to overcome the limitations of parametric studies. They further explain that the cumulative distribution function defines the probability that a random variable takes values less than or equal to x. They conclude that the empirical distribution is the proportion of observations of a random sample which are less than or equal to a value x; it's a step function that jumps by $\frac{1}{n}$ at each data point x_i .(Ayari and Boutahar,2021).

Suppose $X_1, X_{2,\ldots,}X_n$ are i.i.d random variables, the empirical distribution function is defined by:

$$F_n = \frac{1}{n}\sum_{i=1}^{n} \mathbb{I}(X_1 \leq x) \qquad (3.9)$$

Where $\mathbb{I}$ is the indicator function.

For a bivariate dataset we calculate the empirical distribution functions for each of the variables X and Y separately. These EDFs provide non-parametric estimates of the marginals of each variable.

3.5.2Maximum Pseudo-Likelihood Method

Copulas need methods that are used to extract the dependence structure from them, this study will use the MPL method. WANG (2023), both the FML and IFM are parametric estimation methods, which means the performance of algorithms is dependent on the accuracy of estimated parameters. WANG (2023), further explains that to address the potential problem of the misspecification of marginal distributions, Pseudo Maximum Likelihood, a semiparametric estimation, was proposed. This method is the best one to use considering that the study used empirical distribution functions to estimate the marginals. Wang (2023), confirmed that it is similar to the IMF, including two steps to carry out, but instead of estimating the marginal distributions parametrically , it utilizes the empirical distribution of the samples.

Let $\hat{f}_1, k = 1, \ldots\ldots, n$ denote the empirical density of X_k.

$$\sum_{i=1}^{n} log c_\theta [\hat{F}_1 \left(X_{1,i}\right), \ldots\ldots, \hat{F}_d\left(X_{d,i}\right) = \sum_{i=1}^{n} log c_\theta (\hat{U}_{1,i}, \ldots\ldots, \hat{U}_{d,i}) \quad (3.10)$$

Where $c_\theta$is the copula density function and the $\hat{F}_1$ is the empirical CDF calculated earlier on in the methodology.

3.5.4 Dependence measures

Kendall's Tau

Gosh et al. (2022), explained that Kendall's tau measures the amount of concordance present in a bivariate distribution. According to Foresto (2024), Kendall's tau is a non-parametric measure of dependence between the random variables commonly associated to copulas. Further ,studies like Basvi (2024) confirmed the use of this parameter estimation method. In this study, parametric copula estimation and copula parameter estimation from kendall's tau were the methods utilized to fit copulas to data. Because of the link between rank correlation coefficients and copulas and the relative simplicity of the computations, the former method is more attractive, Basvi (2024). Below is the formula for calculating the kendall's tau;

$$\tau c = 4 \iint_0^1 C(u, v) dC(u, v) - 1 \quad (3.11)$$

Where c is the copula associated with the random variables say (x,y).

- U, V are the marginals.

3.6 Model selection

This area of the methodology aims to select the best copula that could be used to represent the dependence structure between stock market performance (DCI) and real interest rates. This study uses the Alkaline information criterion and the Bayesian information criterion. Shabeygi et.al (2025), explained AIC and BIC are the most common criteria used to select the model among the available models. They further explain that the lower AIC or BIC values are generally preferred.

3.6.1 Akaike Information Criterion (AIC)

Alshenawy (2024), the Akaike Information Criterion (AIC) is a well established method used to compare different parametric models based on their fit to the data while penalizing for the number of parameters used. Ashenawy (2024) defined the AIC as :

AIC= 2k-2ln(L) (3.13)

Where;

-k is the number of parameters in the model, and L is the maximum value of the likelihood function for the model.

3.6.2 Bayesian Information Criterion

Shabeygi et al. (2025), BIC seeks to select the model Mi that maximizes the posterior probability. This criterion (BIC) is obtained using the following relationship.

$$BIC = -2\log(L) + 2Klog(N) \quad (3.14)$$

Where

L= maximum likelihood of the fitted copula model

K= number of estimated parameters in the copula

n= number of observations

3.7 Concluding remarks

This section had the methods that will be used to model the dependence between interest rates and stock market performance. An integration of parametric and non-parametric methods was the direction this paper went in. This direction is a way to avoid problems associated with fully parametric methods but it also approaches copulas in a less popular way providing a modern alternative to modelling particularly in Botswana.

CHAPTER 4

ANALYSIS AND DISCUSSION OF FINDINGS

This section will explore the results acquired from the analysis of the real interest rate data and the BSE DCI. Various copulas were employed to extract the dependence structure from the two variables of interest. This was quite rigorous to fully understand just how the two variables move in Botswana's economy. However, this section will showcase just how powerful the copula model is and why it should be used in more statistical studies. The layout of the section will include; the measures of central tendencies including kurtosis and skewness particularly what they inform us in terms of copulas, Q-Q plots to further test for normality, JB test results, simple data visualization, results from different copulas, Kendall's tau as a dependence measure, and finally testing the model selection criteria.

4..1PRELIMINARY ANALYSIS

The following tables and figures will report exploratory analysis of the real interest rates (%) and the DCI (log returns).

Exploratory data analysis Table 1

MEASURE	Real Interest rates (%)	DCI(LOG_returns)
Mean	2.518657	0.00179158
Median	3.5	0.001406
Kurtosis	4.812793	3.687961
Skewness	-1.731173	0.3624089

Table 1 above contains the preliminary analysis of the data before any model is applied to it. It presents the mean and median which shows the general values of the Prime rates and the Log returns of the DCI. To better understand the data, skewness and kurtosis statistics were calculated as well. A skewness of -1.731171 for the real interest rates suggests that the data is highly skewed, it suggests left skewness. According to Hatem et al. (2022), if the skewness is less than -1 or greater than 1 then the data is moderately skewed. This may even suggest some level of asymmetry which supports the need for copula modelling in this study. However, the skewness for the log returns are equal to 0.3624069 which suggests that the data is fairly symmetrical. The kurtosis values are used to assume normality and investigate for tail heaviness. The real interest rates kurtosis is 4.812793, suggesting that the data is moderately leptokurtic (heavy tailed) supporting the need to model dependence using copulas. Kurtosis value of the DCI log returns which was 3.687961 which suggests the log returns are slightly leptokurtic (slight heavy tails), which may suggest that the log returns of the DCI are normal. It is worth noting that we cannot confirm normality based on the Kurtosis alone hence we shall employ other tests as this study unravels.

4.1.2 Q-Q PLOT

Figure 1 below shows the Q-Q plot of DCI log returns. The results suggest a that the DCI log returns follow a normal distribution which is consistent with the results from the table above from the kurtosis and skewness tests. Figure 2 is the Q-Q plot of the Prime real interest rates (%). Unlike the log returns of the DCI, the real interest rates data shows clear deviations from normality, this is rather consistent with the results from the kurtosis and skewness. The plot displays strong left skewness and heavy left tail behavior. Which also supports the decision to model the dependence structure of these two variables using copulas. Other traditional means like regression or linear correlation coefficient may have a difficult time trying to show the dependence.

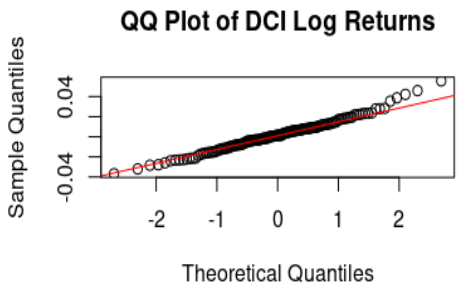

Figure 1

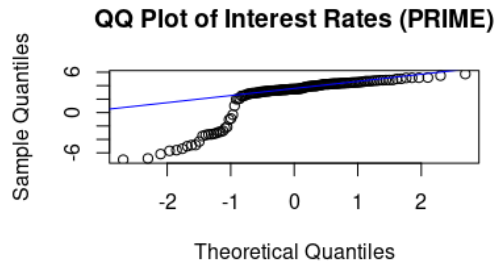

Figure 2

4.1.3Normality test

Below are results for the Jarque-Bera (JB) test to test the data for normality. As mentioned, some copulas cannot fit non normal data, therefore it's best to test for normality to narrow the selection of copulas.

Table 2

Test	Real Interest rates (%)	DCI (log returns)
Jarque Bera (JB)	2.2e-16	0.05147

The Jarque Bera (JB) shows a very low p value for the real interest rates violating the assumption of normality. There is overwhelming evidence that the real interest rates are nonnormal. On the other hand, the p value for the log returns is slightly above 0.05 indicating some normality in the log returns for the DCI. This is consistent with the results provided by the kurtosis. These results, are particularly important in this study as they provide an idea as to which copula might best fit

the data. It may appear that copulas like the gaussian that depend on normality might not be the best fit for the data. However, this will be confirmed by the model selection criteria for copulas.

4.1.4Scatter Plot For Real Interest Rates And Log Returns Of The DCI

The figure below shows a simple scatter plot to investigate if there is any clear relationship between interest rates and stock market performance. The scatter plot does not show any clear relationship between interest rates and DCI. The interest rates are rather clustered around a specific point showing that interest rates may have been constant for a long period of time. This scatter plot does not indicate any linear relationship between the two variables. This supports the need for copulas in this study. Other statistical models like the linear correlation coefficient might have failed to fit such data.

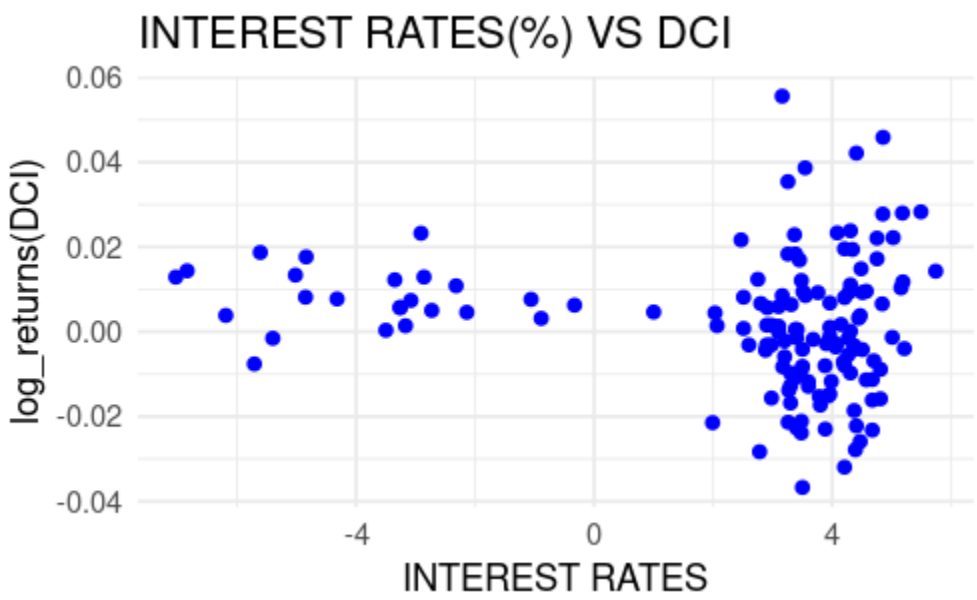

Figure 3

4.2**Marginals Of The Real Interest Rates Vs DCI(log returns)**

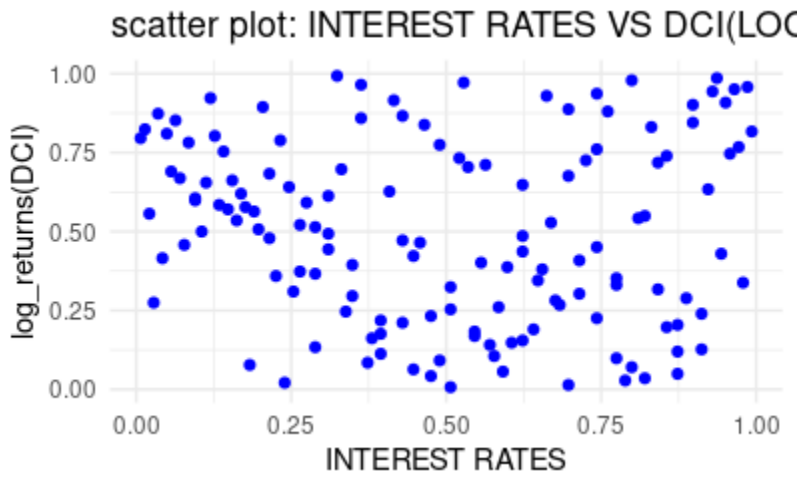

Figure 4

This is a scatter plot of the marginals that were created using the EDFS as the methodology explained. Consistent with the original data of the DCI and real interest rates this shows that there is no clear relationship between real interest rates and stock market performance (DCI).

4.3Copula Parameters And The Kendall's Tau Statistic

Table 3 below shows the Kendall's Tau statistic and table 4 show parameters extracted from four copula models as well as their Kendall's Tau estimates.

The Kendall's tau presented below indicates that there is a weak negative dependence between real interest rates and DCI. These results are consistent with a study by Arendas(2022), the estimated Kendall's tau is -0.4956, suggesting significant negative relationship between observed variables. This kendall's tau suggests that an increase in one variable would lead to a decrease in another variable ,however only weakly.

Table 3

KENDALL'S TAU STATISTIC (τ)
$(\tau) = -0.0700937$

Table 4

COPULA	**Parameter**	**Kendall's tau**
GAUSSIAN	0.1288	-0.02305577
Plackett	0.776992	-0.05631819
FRANK	-0.508	-0.05630127
T copula	-0.03705	0.01955048

4.4Model Selection

To best determine the model that fits the data this paper employed the AIC and the BIC tests. The results are presented below. As an additional measure the log likelihoods from the copulas are added to ensure that the best copulas are chosen.

4.4.1AIC AND THE BIC RESULTS

Table 5

COPULA	AIC VALUES	BIC VALUES	Log likelihood
GAUSSIAN	1.8369194	4.785679	0.08154
FRANK	1.0210951	3.969855	0.4895
PLACKETT	1.0267154	3.975475	0.4866
T cop	3.8365150	9.734035	0.08174

According to Wang (2023), gaussian copula and the T copula are better than the other three copulas based on a larger log-likelihood and smaller AIC and BIC values. Based on the afore statement, the frank is the best fitting model as it has the largest log likelihood and the smallest AIC and BIC values. The tau parameter from the frank which is -0.05673 is really close to the empirical Kendall's tau indicating the copula fits the data well. In conclusion the Frank copula is the most appropriate measure of the negative dependence.

The parameter extracted from the frank copula was a -0.508 and a kendall's tau of -0.05673 which both indicate a negative dependence between real interest rates and DCI. It is worth noting that from the results the gaussian copula and the t copula failed the tests the most. Them having very low likelihoods, and large BIC and AIC value, this is consistent with the previous results of normality. Clemente and Romano (2021) stated that the gaussian copula is the copula of the

multivariate normal distribution, they further explain that the t copula is the copula of the t student distribution. These copulas, assume normality. However, the interest rates showed heavy left tails and failed the normality tests which is consistent with them being the worst fitting copula. Furthermore, those copula model linear dependence (elliptical copulas) but the results from the scatter plot shows no linear relationship between the variable. Therefore, them failing was expected.

4.5 Concluding Remarks

In conclusion, the analysis of data showed that the data was non normal and heavy tailed. This supports the decision to use copulas. The study suggested that the frank copula was the best fitting model to the data. it captures the dependence of the two variables the best. An empirical tau of -0.0701 indicated weak negative dependence between the two variables. The frank copula parameter which was -0.508 and the corresponding tau -0.05630127 also suggest that there is a weak negative dependence between real interest rates and the domestic company index. The closeness of the tau of the frank model to that of the empirical tau indicates that the frank is an appropriate fit. This suggest that when one variable increase the other may decrease however only weakly. The inverse relationship is not strong. This negative dependence is consistent with other studies, Alzoubi (2022) stock market performance is highly elastic to real economic activity; a 1 percent increase (decrease) leads to almost 4 percent increase (decrease) in stock prices. They further explain that the central bank should be cautious before raising the interest rate because of its strong negative association with the stock market. Although in Botswana the relationship is weak , there still some negative association.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter provides a summary of the finding of this study. Finding drawn from the analysis based on the research objectives that were crafted at the beginning of this study. It will also provide the recommendations to other authors and researchers on how to improve this study, moreover even give advice to the government on macro-economic issues that may have an impact on the stock market.

5.2 Research objective and conclusion

The main objective of this study was to model dependence between interest rates and stock market performance using a copula. This study settled on the frank copula which indicated a weak negative dependence between interest rates and stock market performance. The Kendall's tau also indicated that there is a weak negative dependence. This study indicates that an increase in the level of one variable say interest rates would lead to decrease in DCI stock returns however it also provides the strength which is weakly. Most authors describe such a relationship as insignificant or not entirely important.

5.3 Conclusion

Rapaeye (2017), who conducted a study on Banks, stock market and economic growth in Botswana concluded that interest rates by virtue of them being a cost of borrowing, are expected to have a negative impact on the stock returns but the findings of the study show that they have an insignificant relationship with stock returns. Rapaeye (2017), indicated that DCI returns are weakly positively correlated to interest rates and money supply with correlation coefficients of 0.039 and 0.091 respectively. It's interesting how this study found a weak positive relationship and concluded that the relationship is insignificant. My study contradicts this conclusion by finding a weak negative dependence, it's evident that as years pass this relationship may change. The afore mentioned study was conducted in 2018, 7 years ago before the hike in inflation due to

covid 19. My study that's after covid, produced differing results cementing its significance. It shows that this relationship changes overtime hinting a need for regular updating.

5.4 Recommendations

It is recommended that the government take proper caution when dealing with interest rates. Although this study concluded that the relationship between interest rates and stock market performance is weak, there is some shift in the relationship. As stated above, one study proved that the there was a weak positive relationship and now this study is concluding that there is a weak negative relationship. This proved that the changes in interest rates may have some significant effect on the stock market as time goes on. I also recommend that other authors apply time series methodologies to incorporate some forecasting element to this study. Other researchers could include other economic variables such as inflation to determine which variable has a larger dependence on the stock market still applying copulas. For this topic, in particular researchers could go a completely different route by conducting a parametric study. Doing so will allow for the use of the CvM tests so as to have some level of significance for the copula.

5.4.1 Conclusion

This study showed that the datasets revealed some asymmetry, and nonnormality. These results showed that using traditional means of analysis like the pearson correlation coefficient and linear regression methods would have failed to show the relationship between interest rates and stock market performance. Cementing the need for copulas in this study. The main results is that there was some weak negative relationship between interest rates and stock market performance. The strength of the relationship indicates that the relationship is not that significant. However, it contradicts other older studies meaning that this relationship is prone to change over the years. The relationship should be investigated using other models and updated every few years.

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